



SamaRasa

AI-Powered Companion and Health Management System

Interface Description & User Manual

SSW3001 Software Engineering

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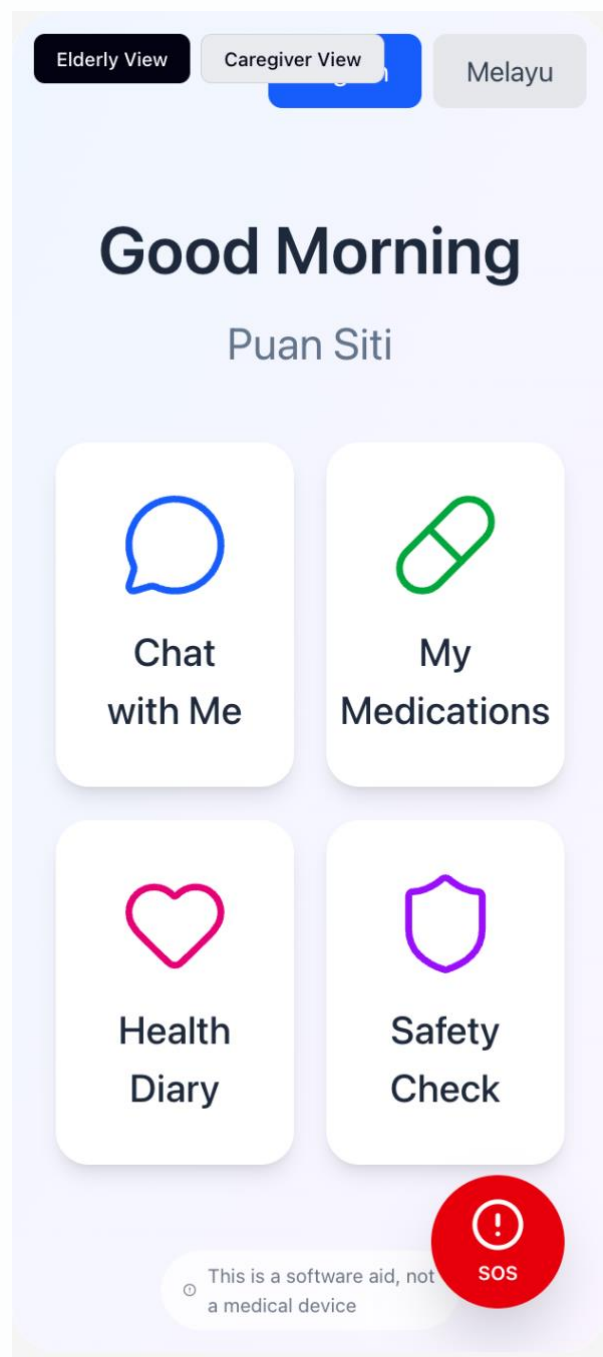
Interface Description

This section provides a detailed description of all interface elements and their functions in the SamaRasa system. The system consists of two main user interfaces: the Elderly User Interface (tablet application) and the Caregiver Dashboard Interface (web portal).

Elderly User Interface

1. Home Screen

The Home Screen is the first interface displayed when an elderly user opens the application. It is designed with simplicity and ease of use in mind.



Interface Element	Function Description
Greeting Message	Displays time-based greetings such as "Good Morning" or "Selamat Pagi"
User Name	Shows the current user's name, e.g., Puan Siti
Feature Card Area	2x2 grid layout containing four main function entry points
Chat with Me	Opens the AI companion chat interface
My Medications	Displays medication schedule and reminders
Health Diary	Opens mood logging and health status recording
Safety Check	Opens the fall risk assessment questionnaire
SOS Button	Fixed red emergency button at the bottom-right corner
Language Toggle	Top buttons to switch between English and Melayu

The Home Screen uses high-contrast colors (meeting WCAG AAA standards), with font sizes no smaller than 24px and touch targets no smaller than 48x48 pixels, ensuring elderly users can easily identify and operate the interface.

2. AI Chat Interface

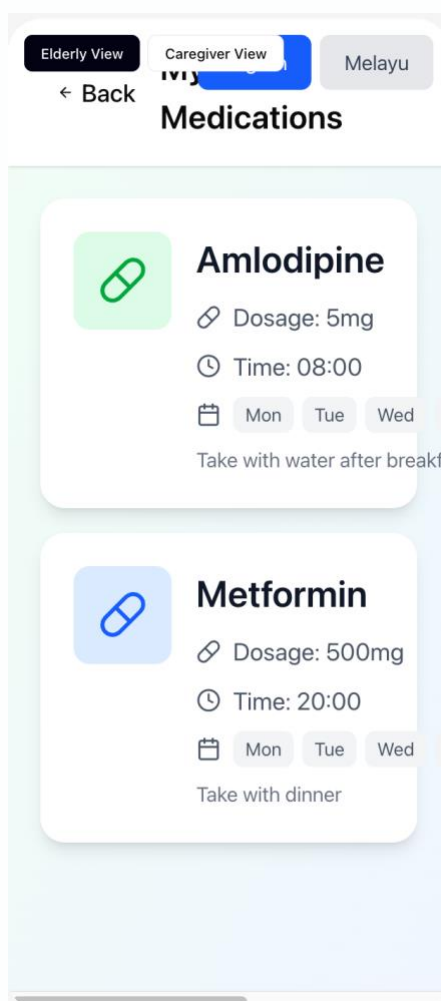
The AI Chat Interface provides voice interaction functionality, allowing elderly users to have natural conversations with the AI companion.



Interface Element	Function Description
Back Button	Returns to the Home Screen from the top-left corner
Title	"Chat with Me" identifies the current function
AI Message Bubble	Displays AI greetings or responses, e.g., "How are you feeling today?"
Voice Input Button	Large blue circular button at the bottom with microphone icon
Voice Animation	Pulsing waveform animation displayed during recording
Status Indicator	Shows statuses like "Listening", "Thinking..."

3. Medication Management Interface

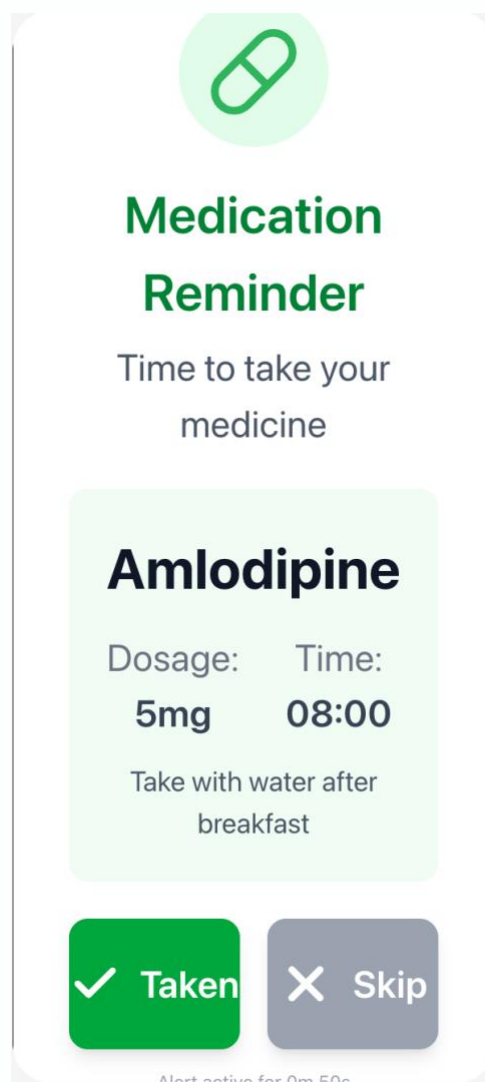
The Medication Management Interface displays all medication schedules for the elderly user in card format for easy viewing.



Interface Element	Function Description
Pill Icon	Colored pill icon at the top of each medication card
Medication Name	Such as Amlodipine, Metformin, etc.
Dosage Information	Displays medication dosage, e.g., 5mg, 500mg
Medication Time	Shows daily medication time, e.g., 08:00, 20:00
Medication Days	Date tags showing which days to take (Mon, Tue, etc.)
Instructions	Shows medication notes, e.g., "Take with water after breakfast"

4. Medication Reminder Interface

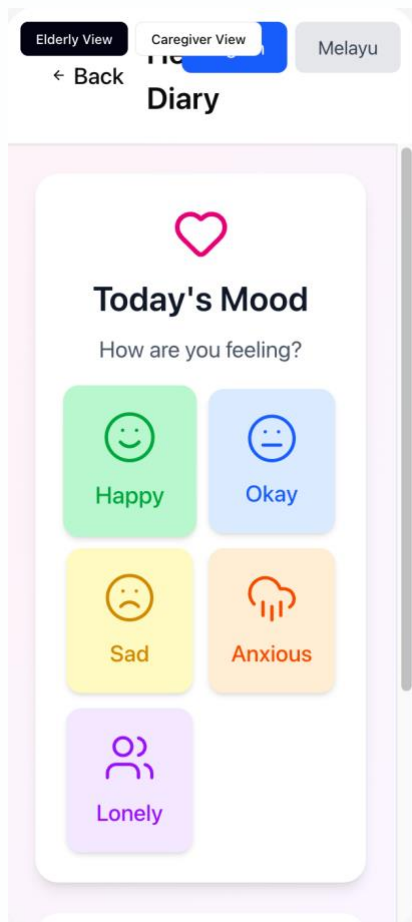
The Medication Reminder is a full-screen popup interface that ensures elderly users do not miss their medication times.



Interface Element	Function Description
Large Pill Icon	Large pill icon displayed at the center of the screen
Reminder Title	"Medication Reminder" identifier
Reminder Subtitle	"Time to take your medicine" prompt
Medication Details Card	Displays medication name, dosage, time, and instructions
Taken Button	Large green button indicating medication has been taken
Skip Button	Gray button to skip this medication dose
Timer	Shows how long the reminder has been active, e.g., "0m 50s"

5. Health Diary Interface

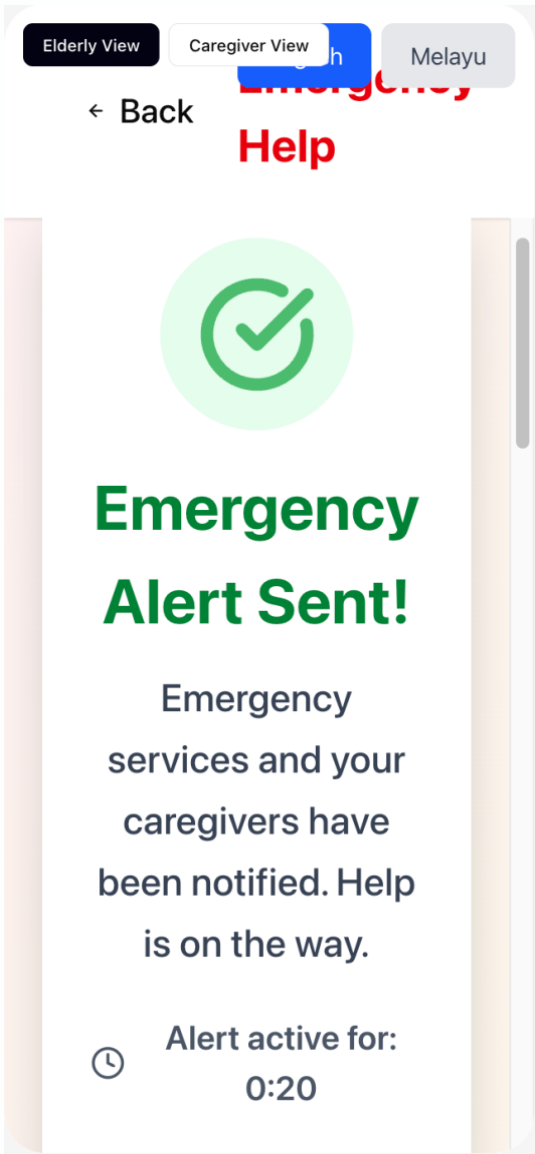
The Health Diary Interface allows elderly users to record their daily mood, helping track emotional changes over time.



Interface Element	Function Description
Title	"Health Diary" identifier
Mood Recording Card	"Today's Mood - How are you feeling?" prompt
Mood Options	2x2+1 grid layout of emoji mood choices
Happy	Happy mood (green)
Okay	Neutral mood (blue)
Sad	Sad mood (yellow)
Anxious	Anxious mood (orange)
Lonely	Lonely mood (purple)

6. Emergency SOS Interface

The SOS Interface is triggered during emergencies to send alert information to caregivers.

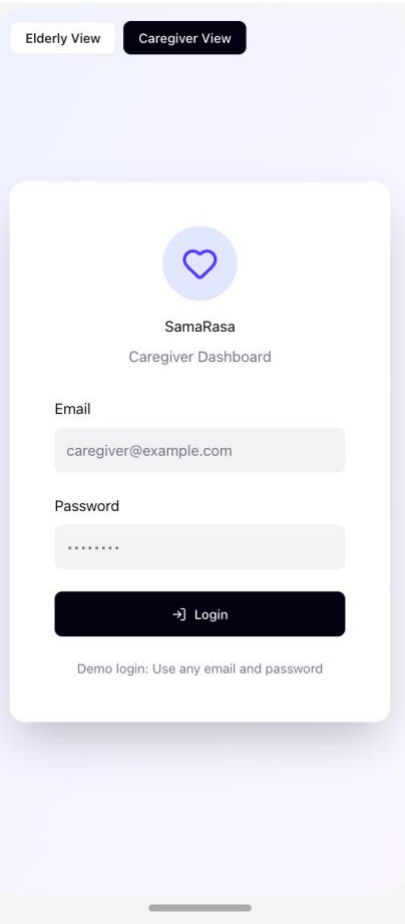


Interface Element	Function Description
Checkmark Icon	Large green checkmark indicating alert has been sent
Success Title	Red text displaying "Emergency Alert Sent!"
Confirmation Message	Informs user that emergency services and caregivers have been notified
Timer	Shows how long the alert has been active, e.g., "0:20"

Caregiver Dashboard Interface

1. Login Interface

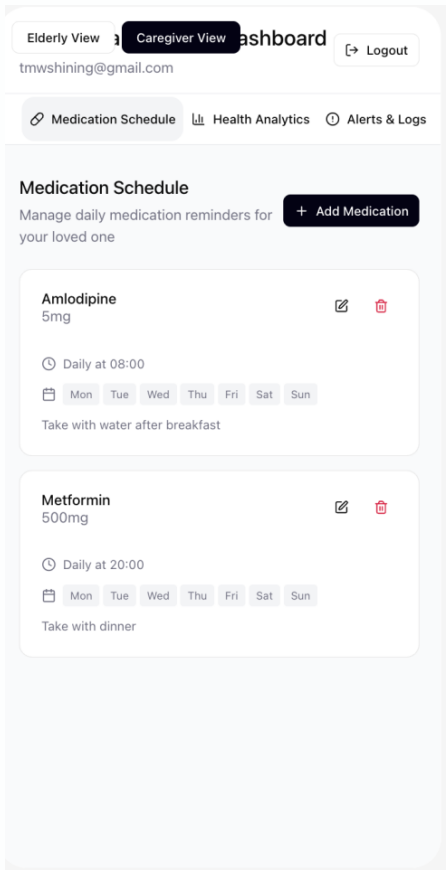
Caregivers access the system through the Login Interface, which requires a valid email and password.



Interface Element	Function Description
Logo	SamaRasa brand identifier with heart icon
Subtitle	"Caregiver Dashboard" indicating the current system
Email Input Field	Enter caregiver email address
Password Input Field	Enter login password (displayed as dots)
Login Button	"Login" button to submit credentials
Helper Text	Demo login instructions for testing purposes

2. Medication Schedule Interface

The Medication Schedule is the main page of the Caregiver Dashboard, used for managing elderly users' medication schedules.

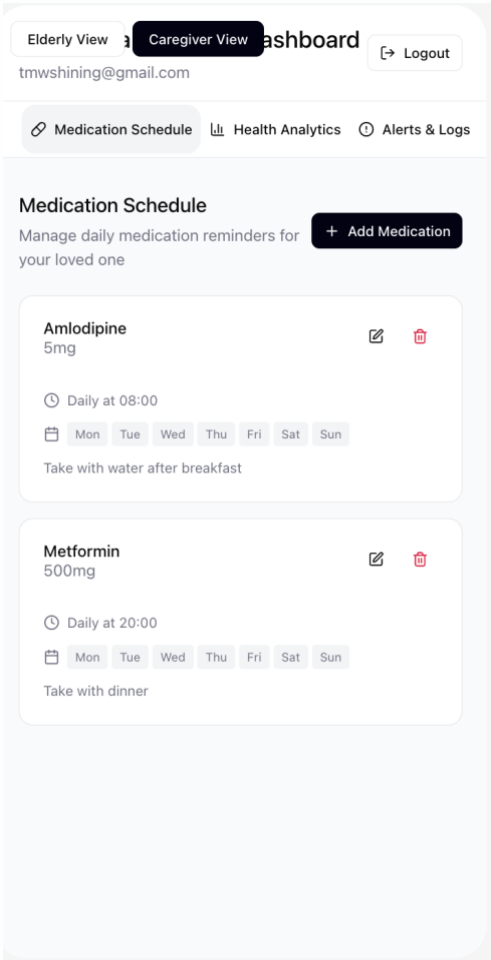


Interface Element	Function Description
Header Title	"SamaRasa Dashboard" brand identifier
User Info	Displays current logged-in caregiver email
Logout Button	Logout button at the top-right corner
Tab Navigation	Medication Schedule / Health Analytics / Alerts & Logs
Page Title	"Medication Schedule" identifier
Page Subtitle	Function description: "Manage daily medication reminders"
Add Button	"+ Add Medication" to add new medications
Medication Cards	Display detailed information for each medication

Interface Element	Function Description
Edit/Delete Buttons	Action buttons at the top-right of each card

3. Health Analytics Interface

The Health Analytics Interface displays charts showing medication adherence and mood trends for the elderly user.

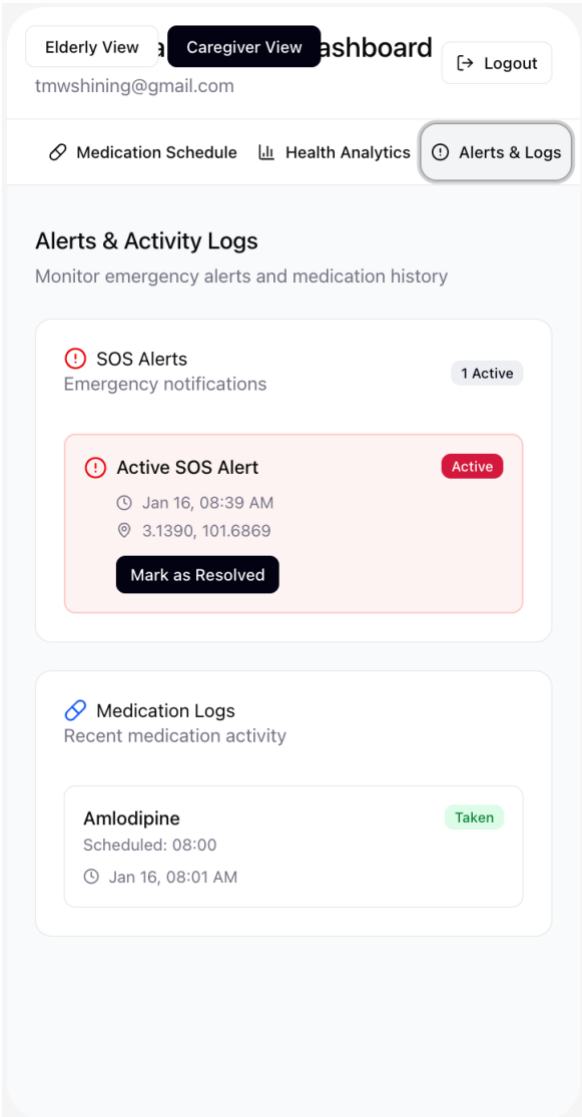


Interface Element	Function Description
Medication Tracking Chart	Bar chart showing daily medication status
Legend	Missed (red) / Skipped (yellow) / Taken (green)
X-Axis	Displays date range (e.g., 12/20 - 1/16)
Mood Trend Chart	Line graph showing 14-day mood changes

Interface Element	Function Description
Y-Axis	Mood score range from 0-10
Chart Titles	"Daily tracking of medication intake" and "Mood Trend"

4. Alerts & Logs Interface

The Alerts & Logs Interface displays emergency alerts and medication records, allowing caregivers to quickly understand any abnormal situations.



Interface Element	Function Description
SOS Alerts Section	"SOS Alerts - Emergency notifications" header
Active Alert Card	Displays currently active SOS alerts

Interface Element	Function Description
Status Badge	"Active" red badge indicating alert status
Trigger Time	Shows when the alert was triggered, e.g., Jan 16, 08:39 AM
GPS Location	Displays trigger coordinates (3.1390, 101.6869)
Resolve Button	"Mark as Resolved" to mark alert as handled
Medication Logs Section	"Medication Logs - Recent medication activity"
Log Entries	Detailed information for each medication record
Status Badge	"Taken" green badge indicating medication taken

User Manual

System Overview

SamaRasa is an AI-powered companion and health management system designed specifically for elderly Malaysians. The system supports bilingual interaction in English and Bahasa Malaysia, with a voice-first design philosophy, making it particularly suitable for elderly users with limited digital literacy.

The system includes six main features:

1. **AI Chat Companion:** Provides daily greetings and emotional support
2. **Smart Medication Reminder:** Scheduled reminders with medication tracking
3. **Health Diary:** Records daily mood and health status
4. **Emergency SOS Alert:** One-touch emergency call for help
5. **Fall Risk Assessment:** Periodic fall risk evaluation
6. **Caregiver Dashboard:** Remote monitoring and management

Quick Start Guide

Quick Start for Elderly Users

1. Open the application and wait for the AI greeting to finish playing 2. Tap "Chat with Me" card to chat with AI; hold the microphone button to speak 3. Tap "My Medications" to view today's medication schedule 4. When medication time arrives, the system will automatically display a full-screen reminder 5. Tap "Taken" to confirm medication taken, or tap "Skip" to skip 6. In case of emergency, long-press the SOS button for 3 seconds or double-tap

Quick Start for Caregivers

1. Access the Caregiver Dashboard URL 2. Enter email and password to log in 3. On the "Medication Schedule" page, click "+ Add Medication" to add new medications 4. Fill in medication name, dosage, time, and instructions 5. Switch to "Health Analytics" to view medication adherence and mood trends 6. On the "Alerts & Logs" page, view emergency alerts and medication records

Feature Instructions

AI Chat Feature

The AI Chat feature is based on voice interaction and supports natural conversation. The system performs sentiment analysis on conversation content and automatically records the user's mood status. If keywords like "sad" or "dizzy" are detected, the AI will provide care and suggestions.

Medication Reminder Feature

Medication reminders are automatically triggered at the scheduled time, playing voice alerts and displaying a full-screen interface. If there is no response within 15 minutes, the system will automatically mark the status as "Missed" and send a notification to the caregiver.

SOS Emergency Alert

The SOS button has an anti-accidental trigger mechanism: it requires a 3-second long press or double-tap to activate. After activation, there is a 5-second countdown during which it can be cancelled. When the countdown ends, the system automatically sends an alert containing the user ID, time, and GPS location to the caregiver.

Frequently Asked Questions

Question	Answer
How do I switch languages?	Tap [English] or [Melayu] buttons at the top of the Home Screen
What if medication reminder doesn't respond?	Check if device volume is on and network connection is normal
How do I cancel an accidental SOS trigger?	Tap the "Cancel" button during the 5-second countdown
How do I add new medications?	Caregivers click "+ Add Medication" button in the Dashboard
What if medication records are inaccurate?	Contact caregivers to modify records in the Dashboard
Which browsers are supported?	Chrome 90+, Safari 14+, and other modern browsers
What device do I need?	Recommended: 10-inch or larger tablet with microphone and speaker

For other issues, please contact the system administrator or refer to the complete Software Requirements Specification (SRS) and Software Design Document (SDD) for more technical information.



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